

SQL Query Analysis

Grad Café applicant dataset | 30,384 imported records | Results supplied from the verified console run

Method note: The original downloaded university field is stored in the supplementary applicant_original_universities table, joined to the required 15-column applicants table by p_id. The SQL below is reproduced from the uploaded query_data.py, with its shared query conditions expanded for readability.

Question 1

Question

How many entries in the database are from applicants who applied for Fall 2026?

Final result

Fall 2026 applicant count: 29,902

SQL query used

```
SELECT COUNT(*)  
FROM applicants  
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

Explanation

Counts rows whose term, after trimming spaces and ignoring case, is Fall 2026.

Question 2

Question

Among entries that provide a nationality classification, what percentage are international students?

Final result

Percent international: 46.53%

SQL query used

```
SELECT
100.0 * COUNT(*) FILTER (
WHERE LOWER(TRIM(us_or_international)) = 'international'
)
/ NULLIF(
COUNT(*) FILTER (
WHERE NULLIF(TRIM(us_or_international), '') IS NOT NULL
),
0
)
FROM applicants;
```

Explanation

Counts International entries and divides by all entries with a nonblank nationality classification. American and Other remain in the denominator; blank and NULL classifications do not.

Question 3

Question

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Final result

Average GPA: 3.77

Average GRE Quantitative: 165.88

Average GRE Verbal: 160.80

Average GRE Analytical Writing: 4.36

SQL query used

```
SELECT
  AVG(gpa) FILTER (
    WHERE gpa BETWEEN 0 AND 4
  ) AS average_gpa,

  AVG(gre) FILTER (
    WHERE gre BETWEEN 130 AND 170
  ) AS average_gre_quantitative,

  AVG(gre_v) FILTER (
    WHERE gre_v BETWEEN 130 AND 170
  ) AS average_gre_verbal,

  AVG(gre_aw) FILTER (
    WHERE gre_aw BETWEEN 0 AND 6
  ) AS average_gre_analytical_writing

FROM applicants;
```

Explanation

Computes each average independently, excluding missing values and filtering GPA to 0–4, GRE Quantitative and Verbal to 130–170, and Analytical Writing to 0–6. The GRE field includes combined scores and other out-of-range values, so these are filtered averages rather than averages of all nonmissing values. GPA values on other grading scales are excluded.

Question 4

Question

What is the average GPA of American applicants who applied for Fall 2026?

Final result

Average GPA of American Fall 2026 applicants: 3.79

SQL query used

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(us_or_international)) = 'american'
AND gpa IS NOT NULL;
```

Explanation

Restricts rows to Fall 2026, American classification, and reported GPA, then computes the mean. Unlike Question 3, the current SQL does not impose a 0–4 GPA range.

Question 5

Question

What percentage of Fall 2025 entries are acceptances?

Final result

Fall 2025 acceptance percentage: 47.96%

SQL query used

```
SELECT
100.0 * COUNT(*) FILTER (
WHERE LOWER(TRIM(status)) LIKE 'accept%'
)
/ NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

Explanation

Among all Fall 2025 entries, divides the count with status beginning with “accept” by the total number of Fall 2025 entries.

Question 6

Question

What is the average GPA of accepted applicants who applied for Fall 2026?

Final result

Average GPA of accepted Fall 2026 applicants: 3.78

SQL query used

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accept%'
AND gpa IS NOT NULL;
```

Explanation

Restricts rows to Fall 2026 acceptances with a reported GPA, then computes the mean. Unlike Question 3, the current SQL does not impose a 0–4 GPA range.

Question 7

Question

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science, using original downloaded fields?

Final result

JHU Computer Science master's applicant count: 8

SQL query used

```
SELECT COUNT(*)
FROM applicants AS a
JOIN applicant_original_universities AS u
ON a.p_id = u.p_id
WHERE (
  u.university ~* 'johns[:space:]*hopkins'
  OR u.university ~* '^(^[:alnum:])*jhu(^[:alnum:])|$',
)
AND (
  a.program ~* 'computer[:space:]*science'
  OR a.program ~* '^(^[:alnum:])*cs(^[:alnum:])|$',
)
AND (
  LOWER(TRIM(a.degree)) IN (
    'masters', 'master', 'master's',
    'ms', 'm.s.', 'msc', 'm.sc.'
  )
  OR LOWER(TRIM(a.degree)) LIKE 'master%'
);
```

Explanation

Joins the original-university lookup table to applicants by applicant ID, then matches Johns Hopkins/JHU, Computer Science/CS, and master's degree labels.

Question 8

Question

How many Fall 2026 acceptances are from Computer Science PhD applicants to Georgetown, MIT, Stanford, or Carnegie Mellon, using original downloaded fields?

Final result

Original-field count: 28

SQL query used

```
SELECT COUNT(*)
FROM applicants AS a
JOIN applicant_original_universities AS u
ON a.p_id = u.p_id
WHERE LOWER(TRIM(a.term)) = 'fall 2026'
AND LOWER(TRIM(a.status)) LIKE 'accept%'
AND (
  LOWER(TRIM(a.degree)) IN (
    'phd', 'ph.d.', 'ph.d',
    'doctorate', 'doctoral'
  )
  OR LOWER(TRIM(a.degree)) LIKE 'doctor%'
)
AND (
  a.program ~* 'computer[[:space:]]*science'
  OR a.program ~* '^(^[:alnum:]]cs(^[:alnum:]]|$)'
)
AND
(
  u.university ~* 'georgetown'
  OR u.university ~* 'massachusetts[[:space:]]+institute[[:space:]]+of[[:space:]]+technology'
  OR u.university ~* '^(^[:alnum:]]mit(^[:alnum:]]|$)'
  OR u.university ~* 'stanford'
  OR u.university ~* 'carnegie[[:space:]]+mellon'
  OR u.university ~* '^(^[:alnum:]]cmu(^[:alnum:]]|$)'
)
;
```

Explanation

Joins original university names to applicants and applies Fall 2026, accepted status, PhD, Computer Science, and the four-university matching conditions together.

Question 9

Question

Repeat Question 8 using the LLM-generated program and university fields. Report both counts and their difference.

Final result

Original-field count: 28

LLM-field count: 24

Difference (LLM minus original): -4

SQL query used

```
SELECT COUNT(*)
FROM applicants
WHERE
  LOWER(TRIM(term)) = 'fall 2026'

AND
  LOWER(TRIM(status)) LIKE 'accept%'

AND
  (
    LOWER(TRIM(degree)) IN (
      'phd',
      'ph.d.',
      'ph.d',
      'doctorate',
      'doctoral'
    )
    OR LOWER(TRIM(degree)) LIKE 'doctor%'
  )

AND
  (
    llm_generated_program ~* 'computer[:space:]*science'
    OR llm_generated_program ~* '^(^[:alnum:]]cs(^[:alnum:]]|$)'
  )

AND
  (
    llm_generated_university ~* 'georgetown'
    OR llm_generated_university ~*
      'massachusetts[:space:]+institute[:space:]+of[:space:]+technology'
    OR llm_generated_university ~* '^(^[:alnum:]]mit(^[:alnum:]]|$)'
    OR llm_generated_university ~* 'stanford'
    OR llm_generated_university ~* 'carnegie[:space:]+mellon'
    OR llm_generated_university ~* '^(^[:alnum:]]cmu(^[:alnum:]]|$)'
  )
;
```

Explanation

Keeps the original term, degree, and admission-status filters but matches program and university using the LLM-generated fields. The difference is $24 - 28 = -4$. This is a difference in totals, not proof that exactly four individual records were misclassified.

Comparison note: Differences can arise from normalization or interpretation of program and university names in the LLM-generated fields. The aggregate counts alone do not establish which individual records changed or why.

Original Question 1

Question

Among Fall 2026 applicants classified as American or International, what percentage of each group received an acceptance?

Final result

American: 5,959 accepted out of 15,553 applicants (38.31%)

International: 4,755 accepted out of 13,712 applicants (34.68%)

SQL query used

```
SELECT
  TRIM(us_or_international) AS nationality,
  COUNT(*) AS applicant_count,
  COUNT(*) FILTER (
    WHERE LOWER(TRIM(status)) LIKE 'accept%'
  ) AS accepted_count,
  100.0 * COUNT(*) FILTER (
    WHERE LOWER(TRIM(status)) LIKE 'accept%'
  ) / NULLIF(COUNT(*), 0) AS acceptance_percentage
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(us_or_international))
  IN ('american', 'international')
GROUP BY TRIM(us_or_international)
ORDER BY nationality;
```

Explanation

Filters to Fall 2026 and the two named classifications, groups by classification, and computes accepted counts and within-group percentages. These are percentages of entries in this dataset, not university-wide admission rates.

Original Question 2

Question

Which five universities have the most Fall 2026 applicant entries in the database?

Final result

Stanford University: 737

University of California, Berkeley: 598

Yale University: 580

Princeton University: 563

University of Washington: 543

SQL query used

```
SELECT
  TRIM(u.university) AS university,
  COUNT(*) AS applicant_count
FROM applicants AS a
JOIN applicant_original_universities AS u
  ON a.p_id = u.p_id
WHERE LOWER(TRIM(a.term)) = 'fall 2026'
AND NULLIF(TRIM(u.university), '') IS NOT NULL
GROUP BY TRIM(u.university)
ORDER BY applicant_count DESC, university ASC
LIMIT 5;
```

Explanation

Joins original university names to applicants, filters to Fall 2026 and nonblank university names, groups by exact trimmed university text, and returns the five largest counts. Variants of a university name may be counted separately.